Homework Assignment 1

Math 292, Spring 2026
Rutgers University

Jeffrey Lu

February 17, 2026

Question 4.8

Suppose a is an integer. Assuming $5 \mid 2a$, there exists an integer k such that

$$2a = 5k.$$

Multiplying both sides by 3 and rearranging we have that

$$\begin{aligned} 6a &= 15k \\ a &= 15k - 5a \\ a &= 5(3k - a) \\ a &= 5w \mid \text{where w = 3k-a.} \end{aligned}$$

And, w must be an integer because it is the linear combination of integers. Thus, by the definition of divisibility, we have

$$5 \mid a.$$

Therefore, we can conclude that if $5 \mid 2a$, then $5 \mid a$. $\square$

Question 4.12

Suppose $x \in \mathbb{R}$ and $0 < x < 4$. We let a = (x-2), where a is a real number because it is the linear combination of real numbers. Because the square of a real number must be nonnegative, we have that

$$a^2 = (x-2)^2 \geq 0.$$

Expanding this we get

$$x^2 - 4x + 4 \geq 0.$$

Then, we divide through by -1, flipping the inequality and resulting in

$$\begin{aligned} -x^2 + 4x - 4 &\leq 0 \\ -x^2 + 4x &\leq 4 \\ x(4-x) &\leq 4 \end{aligned}$$

Recall that $0 < x < 4$, which gives us these two facts

1. $x > 0$
2. $4 - x > 0$

We let $b = 4 - x$ and $c = x * b = x(4 - x)$. Since b and x are both positive, c being the product of two positive real numbers must also be positive.

Thus, we can safely divide through by x(4-x), giving us

$$1 \leq \frac{4}{x(4-x)}.$$

Therefore we conclude that if $0 < x < 4$, then $\frac{4}{x(4-x)} \geq 1$. $\square$

Question 4.14

Suppose $n \in \mathbb{Z}$. Then, we have the two possible cases. n could either be a odd integer or a even integer.

Case 1: n is even. If n is an even integer, we can rewrite it as $n = 2x$ for some integer x. Substituting this back into our expression we now have

$$\begin{aligned} 5n^2 + 3n + 7 &= 5(2x^2) + 3(2x) + 7 \\ &= 20x^2 + 6x + 7 \\ &= 20x^2 + 6x + 6 + 1 \\ &= 2(10x^2 + 3x + 3) + 1. \end{aligned}$$

Let $j = 10x^2 + 3x + 3$. Because j is the linear combination of integers, j is also an integer. Hence, our expression reduces to

$$5n^2 + 3n + 7 = 2j + 1, \text{ where } j \in \mathbb{Z}$$

Therefore, by definition of odd, we can conclude that $5n^2 + 3n + 7$ is odd.

Case 2: n is odd. If n is an odd integer, we can rewrite it as $n = 2x + 1$ for some integer x. Substituting this back into our expression we now have

$$\begin{aligned} 5n^2 + 3n + 7 &= 5(2x+1)^2 + 3(2x+1) + 7 \\ &= 20x^2 + 20x + 5 + 6x + 3 + 7 \\ &= 20x^2 + 26x + 12 + 1 \\ &= 2(10x^2 + 13x + 6) + 1. \end{aligned}$$

Let $j = 10x^2+13x+6$. Because j is the linear combination of integers, j is also an integer. Hence, our expression reduces to

$$5n^2 + 3n + 7 = 2j + 1, \text{ where } j \in \mathbb{Z}.$$

Therefore, by definition of odd, we can conclude that $5n^2 + 3n + 7$ is odd.

In both cases, we have shown that the expression turns out to be odd. Therefore, we conclude that if $n \in \mathbb{Z}$, then $5n^2 + 3n + 7$ is odd. □

Question 5.6

Using the contrapositive, we have that given a real number x, suppose that $x \leq -1$, then we must prove that $x^3 - x \leq 0$.

Adding and subtracting 1 from x, we get

$$x - 1 \leq -2$$
$$x + 1 \leq 0.$$

Multiplying x by x-1, which is strictly negative, flips the sign, giving us

$$x(x - 1) \geq 2.$$

Then, multiplying by x + 1, which is non-positive, we flip the sign again

$$x(x - 1)(x + 1) \leq 0.$$

Combining the terms, we have

$$x^3 - x \leq 0.$$

Therefore, we conclude that if if $x^3 - x > 0$, then $x > -1$. □

Question 18a

Suppose $x < y$. Rearranging, we get

$$y - x > 0$$

Now, consider $x^2 + xy + y^2$, completing the square, we have

$$x^2 + xy + y^2 = (x + \frac{y}{2})^2 + \frac{3}{4}y^2$$

Let $a = (x + \frac{y}{2})$.

We know a must be a real number since any linear combination of reals is still a real number. Thus, we have

$$x^2 + xy + y^2 = a^2 + \frac{3}{4}y^2.$$

Any real number squared must be non-negative and we have that x and y can't both be zero from the original inequality. Then, we have two cases either $a = 0$ then $y^2 > 0$, either $a \neq 0$ then $a^2 = 0$, thus we have

$$x^2 + xy + y^2 = a^2 + \frac{3}{4}y^2 > 0.$$

Next, we multiply both sides by $(y - x)$, which we know is strictly positive and won't flip the inequality

$$(y - x)(x^2 + xy + y^2) > 0$$

Distributing the terms, we have

$$\begin{aligned} y(y^2 + xy + x^2) - x(y^2 + xy + x^2) &> 0 \\ y^3 + xy^2 + x^2y - xy^2 - x^2y - x^3 &> 0 \\ y^3 - x^3 &> 0 \end{aligned}$$

Adding x^3 on both sides, we get

$$y^3 > x^3$$

Therefore, we can conclude that if $x < y$, then $x^3 < y^3$. □

Question 29a

Let x be a real number. Let a = x -1. Then, a must be a real number because it is the linear combination of real numbers. Since the square of a real number is non-negative we have

$$a^2 = (x - 1)^2 \geq 0.$$

Expanding, we get

$$x^2 - 2x + 1 \geq 0.$$

Then, we add $6x + 3$ to both sides

$$x^2 + 4x + 4 \geq 6x + 3.$$

We can now factor the left expression

$$(x + 2)^2 \geq 6x + 3$$

Therefore, we can conclude that for any real number x, we have that $(x + 2)^2 \geq 6x + 3$□